# ArtBench-10 Generative Modeling: A Comparative Study of VAE, GAN, and Diffusion Models

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**Abstract.** Nothing yet.

## 1 Introduction

## 2 Problem Formulation

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### 3.1 Data Processing Pipeline

#### VAE

#### DCGAN

#### Diffusion Models

### 3.2 Implementation Details and Reproducibility

### 3.3 Theoretical Rationale for Hyperparameter Choices

The choice of which hyperparameters to tune and the search space design follows 4 main principles: enable sufficient representational capacity to model ArtBench-10's complexity, stable training dynamics, regularization to prevent overfitting, and computational feasibility under resource constraints.

**Variational Autoencoders (VAE)** For the VAE we tuned latent dimensionality, base channels, learning rate, batch size, and $\beta$ (the weight of the KL divergence term in the loss). Latent dimension is intended to control the capacity of capturing the underlying data distribution, expressing a trade-off between reconstruction fidelity and generalization. Base channels determine the width of the convolutional layers, which affects the model's ability to learn complex features. Learning rate was sampled from a log-uniform distribution to allow exploration across several orders of magnitude, which is crucial for stable convergence. The

$\beta$ parameter was also sampled from a log-uniform distribution to explore a wide range of regularization strengths, consistent with the objective function:

$$\mathcal{L} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - \beta \cdot D_{KL}(q(z|x)||p(z)) \tag{1}$$

where $\beta$ controls the trade-off between reconstruction accuracy and latent space regularization.

**Generative Adversarial Networks (GAN)** For GANs, the search space targeted parameters known to affect training stability and sample diversity. The latent dimensionality and the number of feature maps determine generator and discriminator capacity. We additionally tuned the real-label target used in label smoothing, which can reduce discriminator overconfidence and improve gradient quality early in training. The Adam $\beta_1$ parameter was also tuned because adversarial optimization is sensitive to momentum settings.

**Diffusion Models** For diffusion models, we tuned the denoiser width and depth (`model_channels` and `num_res_blocks`), learning rate, batch size, and the number of diffusion timesteps. The timestep count captures the trade-off between computational cost and generative fidelity: more timesteps generally improve sample quality but increase training and sampling cost. In the latent-diffusion setting, the VAE was kept frozen so that the study isolates the effect of denoiser hyperparameters and avoids conflating them with representation learning in the autoencoder.

## 4 Experimental Design

### 4.1 Hyperparameter Optimization Strategy

We performed hyperparameter optimization with Bayesian optimization (Optuna). To keep the search feasible on standard hardware, tuning was conducted on a development subset comprising 20% of the training data. For each model family, candidate configurations were trained on a training split and selected using performance on a held-out validation split. We minimized the following surrogate objectives: (i) *VAE*: validation reconstruction loss; (ii) *GAN*-based models: validation sum of discriminator and generator losses; (iii) *Diffusion models*: validation denoising loss under a fixed Gaussian noise schedule. The hyperparameter search spaces are summarized in Table 1, and the best configurations found on the studies are reported in Table 2. We used validation losses as surrogates during tuning because FID/KID are computationally expensive and high-variance at the per-trial scale: they require generating large sample sets and repeated feature extraction. In contrast, validation losses are available at each epoch and exhibit lower variance which improves sample efficiency under limited computational resources. We note, however, that GAN losses are far from beeing perfect proxies for perceptual sample quality; therefore, we use this objective only for

hyperparameter selection during tuning and rely on qualitative inspection for the selected final parameters.

**Table 1.** Hyperparameter search spaces used in Bayesian optimization.

| **Model** | **Parameters** | **Values / Range** |
|---|---|---|
| VAE | latent_dim, base_channels, batch_size | $\{16, 32, 64\}$; $\{16, 32, 64\}$; $\{32, 64, 128\}$ |
| | lr, $\beta$ | $[10^{-4}, 3 \times 10^{-3}]$ log-uniform; $[10^{-6}, 10^{-3}]$ log-uniform |
| DCGAN | latent_dim, feature_maps, batch_size | $\{128, 256, 512\}$; $\{32, 64\}$; $\{32, 64, 128\}$ |
| | lr, $\beta_1$, real_label | $[10^{-4}, 5 \times 10^{-4}]$ log-uniform; $[0.3, 0.7]$; $[0.85, 1.0]$ |
| PixelUNet | model_channels, batch_size, num_timesteps | $\{32, 64, 96\}$; $\{16, 32, 64\}$; $\{500, 1000\}$ |
| | lr | $[10^{-4}, 5 \times 10^{-4}]$ log-uniform |
| LatentDenoiser | model_channels, num_res_blocks, batch_size | $\{32, 64, 96\}$; $\{2, 3, 4\}$; $\{16, 32, 64\}$ |
| | lr, num_timesteps | $[10^{-4}, 5 \times 10^{-4}]$ log-uniform; $\{500, 1000\}$ |

### 4.2 Evaluation Protocol

Across the selected configurations (Table 2), the optimization process consistently favored smaller batch sizes. This aligns with the well-documented generalization gap observed in large-batch training [2]. Specifically, the inherent noise in small-batch stochastic gradients serves as a form of implicit regularization, facilitating the escape from sharp local minima and leading the model toward flatter, more robust regions of the loss landscape. Regarding the learning rates, the observed values are relatively low, consistent with the need for stable convergence and stability during training, especially for GANs which are known to be sensitive to hyperparameter settings. Latent dimensions reveal important differences across model families: the VAE favored a much smaller latent space (32) compared to the DCGAN (256), which may reflect the different in the model's architectures: GANs don't pay an explicity penalty (KL divergence) for using larger latent spaces, while VAEs are encouraged to find compact representations. A larger latent space may encourage more diverse samples as long as the generator actually uses it. For the VAE, the optimal $\beta$ value is very low, indicating that the model prioritizes reconstruction fidelity over latent space regularization. For the DCGAN, the optimal $\beta_1$ value is around 0.48, which is close to

the commonly used value of 0.5 [5], suggesting a balance between stability and convergence speed. The real_label parameter being close to 1 (0.9836) indicates that the discriminator benefits from a small anti-overconfidence margin, which can help prevent it from becoming too confident and overpowering the generator during training without weakening the learning signal too much.

**Table 2.** Best configurations obtained.

| Parameter | VAE | DCGAN | PixelUnet | LatentDenoiser |
|---|---|---|---|---|
| Best objective value | 0.02658 | 1.68158 | 0.0358 | 0.2598 |
| Learning rate | 0.0005317 | 0.0001261 | 0.0004748 | 0.0001769 |
| Batch size | 32 | 32 | 16 | 32 |
| Latent dimension | 32 | 256 | 64 | 64 |
| Base channels / feature maps | 64 | 64 | 64 | 64 |
| $\beta$ (VAE) | $1.14 \times 10^{-6}$ | – | – | – |
| $\beta_1$ (Adam, DCGAN) | – | 0.4813 | – | – |
| real_label (DCGAN) | – | 0.9836 | – | – |
| num_timesteps (diffusion) | – | – | 1000 | 1000 |
| num_res_blocks (LatentDenoiser) | – | – | – | 4 |

## 5 Results and Interpretation

### 5.1 Quantitative Results

Table 3 summarizes the quantitative evaluation across model families using 5,000 generated samples and 10 random seeds per model. We report mean ± standard deviation across seeds. Arrows indicate whether lower (↓) or higher (↑) values are preferred. The best value in each metric is highlighted in bold.

**Table 3.** Summary of quantitative results (mean ± std across seeds).

| Model | FID ↓ | KID ↓ | Precision ↑ |
|---|---|---|---|
| VAE | 134.228 ± 0.531 | 0.13136 ± 0.00109 | 0.80548 ± 0.00671 |
| DCGAN | **101.248 ± 0.864** | **0.07664 ± 0.00153** | 0.86400 ± 0.00489 |
| PixelUNet | 116.933 ± 0.708 | 0.09972 ± 0.00107 | **0.90804 ± 0.00512** |
| LatentDenoiser | 119.271 ± 0.416 | 0.10068 ± 0.00122 | 0.78472 ± 0.00466 |

Overall, DCGAN achieves the best performence in terms of global distributional alignment obtaining the lowest FID (101.248 ± 0.864) and KID (0.07664 ±

0.00153) scores. This suggests that the DCGAN is more effective at capturing the overall data distribution compared to the other models. The VAE, on the other hand, has the worst performance in terms of FID and KID, which is consistent with the known limitations of VAEs in generating overly smooth reconstructions due to the variational objective, which can lead to blurry samples and less realistic outputs. PixelUNet achieves the highest precision ($0.90804 \pm 0.00512$), indicating that it generates a higher proportion of samples that are close to the real data manifold. However, the higher FID and KID values compared to DCGAN suggest that the improvement in sample realism does not necessarily translate to better overall distributional alignment. They, therefore, generate more realistic samples but with less diversity. The LatentDenoiser achieves intermediate FID and KID values but the lowest precision among the evaluated models. This suggests that while its generated distribution is moderately aligned with the target dataset at a global level, a larger fraction of its individual samples fall outside the real data manifold compared to the other architectures. Such behavior may indicate incomplete convergence or reduced stability in the denoising process during sampling. In addition, the small standard deviations across seeds for all models suggest that the results are consistent and not heavily influenced by random initialization or stochastic training dynamics which is a positive sign of the robustness of the training process.

## 6 Conclusion

## Use of External Resources and AI Tools

This project utilized ArtBench-10 (Kaggle). Gemini assisted in LaTeX formatting and analysis verification.

## References

1. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S.: Gans trained by a two time-scale update rule converge to a local nash equilibrium. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 6626–6637 (2017)
2. Keskar, N.S., Mudigere, D., Nocedal, J., Smelyanskiy, M., Tang, P.T.P.: On large-batch training for deep learning: Generalization gap and sharp minima. CoRR **abs/1609.04836** (2016), `http://arxiv.org/abs/1609.04836`
3. Kingma, D.P., Welling, M.: Auto-encoding variational bayes. arXiv preprint arXiv:1312.6114 (2013), `https://arxiv.org/abs/1312.6114`
4. Liao, P., Cheng, X., Sridhar, K., Li, Y., Phillip, I., Torralba, A.: The artbench dataset for fine-grained artistic style understanding and generation. arXiv preprint arXiv:2206.11404 (2022), `https://arxiv.org/abs/2206.11404`
5. Radford, A., Metz, L., Chintala, S.: Unsupervised representation learning with deep convolutional generative adversarial networks. In: International Conference on Learning Representations (ICLR) (2016), `https://arxiv.org/abs/1511.06434`